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SHRI VIDHYABHARATHI MATRIC.HR.SEC.SCHOOL

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COMMON PUBLIC EXAMINATION –MARCH –2024

XII - MATHEMATICS **TENTATIVE ANSWER KEY**

PART – I			
Q.No	**CODE – A**	**CODE – B**	**MARKS**
1.	(a) $\frac{8}{3}$	(a) 45°	1
2.	(d) $\frac{3\pi a^4}{16}$	(d) 8	1
3.	(a) 10	(a) $\frac{8}{3}$	1
4.	(d) 2	(a) 10	1
5.	(d) 8	(d) $\frac{1}{\sqrt{5}}$	1
6.	(a) $[\vec{\alpha}, \vec{\beta}, \vec{\gamma}] = 0$	(c)-4	1
7.	(d) $\frac{d^2y}{dx^2}$ -y=0	(a) (b) $\frac{-q}{r}$	1
8.	(b) y =0	(d) $\frac{3\pi a^4}{16}$	1
9.	(d) $\frac{1}{(x+1)^2}$ dx	(c) (d) adj(AB)=(adjA) (adjB)	1
10.	(a) x^2+y^2	(a) (a) $[\vec{\alpha}, \vec{\beta}, \vec{\gamma}] = 0$	1
11.	(c) 2	(c) (d) $\frac{d^2y}{dx^2}$ -y=0	1
12.	(d) parabola	(c) 2	1
13.	(c) $\begin{bmatrix} 5 & -2 \\ 3 & -1 \end{bmatrix}$	(d) parabola	1
14.	(a) 45°	(d) 2	1
15.	(d) $\frac{1}{\sqrt{5}}$	(c) $\begin{bmatrix} 5 & -2 \\ 3 & -1 \end{bmatrix}$	1
16.	(b) $\frac{-q}{r}$	(a) x^2+y^2	1
17.	(a)0	(b) y =0	1
18.	(c) 2	(a)0	1
19.	(d) adj(AB)=(adjA) (adjB)	(c) 2	1
20.	(c) -4	(d) $\frac{1}{(x+1)^2}$ dx	1

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PART – II

21.	$\sum_{n=1}^{12} i^n = 0$	2
22.	$\alpha^2 + \beta^2 = \frac{-3}{4}$; $\alpha^2\beta^2 = \frac{169}{4}$ $4x^2 + 3x + 169 = 0$	1 1
23.	df=$(2x+3)$dx =0.18	1 1
24.	y=mx ; $\frac{dy}{dx}$=m y = x$\frac{dy}{dx}$	1 1
25.	mean $= \int_1^2 2x(x-1)dx = \frac{5}{3}$	2
26.	$(x+3)^2+(y+4)^2=9$ $x^2 + y^2 + 6x + 8y + 16 = 0$	1 1
27.	$\begin{vmatrix} -1 & 3 \\ 4 & -7 \end{vmatrix} = -5$ ρ(A)=2	1 1
28.	$\int_0^{\frac{\pi}{2}} \sin^{10} x dx = \frac{63\pi}{512}$	2
29.	$\lim_{x\to 1} \frac{x^2-3x+2}{x^2-4x+3} = \lim_{x\to 1} \frac{2x-3}{2x-4}$ $= \frac{1}{2}$	1 1
30.	$\begin{vmatrix} 2 & -1 & 3 \\ 1 & -1 & 0 \\ 3 & -1 & 6 \end{vmatrix} = 0$ ∴The given vectors are coplanar	1 1

PART – III

31.	Let, cot$\boldsymbol{\alpha} = \frac{1}{\sqrt{x^2-1}}$ sec$\boldsymbol{\alpha}$=x ; $\boldsymbol{\alpha}$=sec^{-1}x cot$^{-1}\left(\frac{1}{\sqrt{x^2-1}}\right)$= sec^{-1}x	1 1 1
32.	$y = \frac{-1}{4}(x^2 + 6x + 5)$ $\frac{dy}{dx} = \frac{-1}{4}(2x+6)$ m=-2 Equation of tangent:2x+y+1=0 Equation of normal: x-2y-7=0	1 1 1

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33.	LHS= $\begin{vmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ -1 & 0 & 1 \end{vmatrix} [\vec{a}, \vec{b}, \vec{c}]=0$ $\therefore [\vec{a}-\vec{b}, \vec{b}-\vec{c}, \vec{c}-\vec{a},]=0$	2 1
34.	$u(\lambda x,\lambda y) = \frac{\lambda^{2(x^2+y^2)}}{\sqrt{\lambda}\sqrt{x+y}} = \lambda^{\frac{3}{2}}u(x,y)$ $n=\frac{3}{2}$ By euler's theorem: $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=\frac{3}{2}u$	1 2
35.	x+y=12⟹y=12-x p=xy =x(12-x) p'(x)=12-2x⟹$p'(x)=0$ x=6 p''(x) = -2<0, p is maximum. ∴x=6,y=6	1 1 1
36.	$I=\int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} \frac{1}{1+\sqrt{\tan x}}\,dx$--------1 $I=\int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} \frac{\sqrt{\tan x}}{1+\sqrt{\tan x}}\,dx$--------2 From 1+2 $I=\frac{\pi}{8}$	1 2
37.	$\left(\frac{1+i}{1-i}\right)^3-\left(\frac{1-i}{1+i}\right)^3$ = -i-i = -2i	2 1
38.	$\int \frac{dy}{1+y^2} = \int \frac{dx}{1+x^2}$ $\tan^{-1}y = \tan^{-1}x+c$	1 2
39.	x: 0, 1, 2, 3 f(x): $\frac{1}{8}$, $\frac{3}{8}$, $\frac{3}{8}$, $\frac{1}{8}$	3
40.	|A| = - 1 $\|adj(adjA)\|=\|A\|^{(n-1)^2}=(-1)^4=1$ **Note: Any other method**	1 2
	PART – IV	
41.(a)	Rough Diagram Point of intersection=$\left(\frac{3}{2},\frac{9}{4}\right)$ $m_1=3$; $m_2=-3$ $\theta=\tan^{-1}\left(\frac{3}{4}\right)$	1 1 1 2

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41.(b)

$$\tan^{-1}\left(\frac{\frac{x-1}{x-2}-\frac{x+1}{x+2}}{1-\left(\frac{x-1}{x-2}\right)\left(\frac{x+1}{x+2}\right)}\right)=\frac{\pi}{4}$$ — 1

$$\frac{(x-1)(x+2)+(x+1)(x-2)}{(x-2)(x+2)+(x+1)(x-1}=1$$ — 2

$$2x^2-4=-3 \Longrightarrow X=\pm\frac{1}{\sqrt{2}}$$ — 2

42.(a)

x	2	4	6	8	10
f(x)	$\frac{1}{36}$	$\frac{4}{36}$	$\frac{10}{36}$	$\frac{12}{36}$	$\frac{9}{36}$

— 2

$$F(x)=\begin{cases} 0 \text{ , if } -\infty < x < 2 \\ \frac{1}{36} \quad \text{if } 2 \le x < 4 \\ \frac{5}{36} \quad \text{if } 4 \le x < 6 \\ \frac{15}{36} \quad \text{if } 6 \le x < 8 \\ \frac{27}{36} \quad \text{if } 8 \le x < 10 \\ 1 \text{ if } 10 \le x < \infty \end{cases}$$ — 2

$P(4 \le x < 10)=\frac{13}{18}$ — 1

42.(b)

$$\left(\frac{2z+1}{iz+1}\right)=\frac{2(x+iy)+1}{i(x+iy)+1}$$ — 1

$$\text{Im}\left(\frac{(2x+1)+i2y}{(1-y)+ix}\right)=0$$ — 2

$$\Rightarrow \frac{2y(1-y)-x(2x+1)}{(1-y)^2+x^2}=0$$ — 1

$2x^2+2y^2+x-2y=0.$ — 1

43.(a)

Rough Diagram — 1

$r=\frac{5}{12}h$ — 1

$V=\frac{1}{3}\pi\left(\frac{5}{12}h\right)\left(\frac{5}{12}h\right)h$ — 1

$\frac{dV}{dt}=\frac{25}{144}\pi\times h^2\times\frac{dh}{dt}$ — 1

$\frac{dh}{dt}=\frac{9}{10\pi}$ m/min — 1

43.(b)

Diagram — 1

$\overrightarrow{OP}=\cos\alpha\vec{i}+\sin\alpha\vec{j}$

$\overrightarrow{OQ}=\cos\beta\vec{j}+\sin\beta\vec{j}$ — 1

$$\overrightarrow{OQ}\times\overrightarrow{OP}=\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos\beta & \sin\beta & 0 \\ \cos\alpha & \sin\alpha & 0 \end{vmatrix}$$

$=\hat{k}(\sin\alpha\ \cos\beta-\cos\alpha\sin\beta)$ . . . . .(1) — 1

By definition

$\overrightarrow{OQ}\times\overrightarrow{OP}=|\overrightarrow{OQ}||\overrightarrow{OP}|\sin(\alpha-\beta)\,\hat{k}=\sin(\alpha-\beta)\,\hat{k}$. . . . .(2) — 1

From (1) and (2)

$\sin(\alpha-\beta)=\sin\alpha\ \cos\beta-\cos\alpha\sin\beta$ — 1

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44.(a)	Rough Diagram $x^2 = -4ay$ $a = \frac{9}{10}$ It's passes through,$(x_1,-7.5)$ $x^2 = -4\left(\frac{9}{10}\right)y$ $x_1 = 3\sqrt{3}$mt.	1 1 1 1 1
44.(b)	$P = \frac{1}{x}$, $Q = \sin x$ I.F= $e^{\int p\,dx} = x$ $xy = \int x\sin x\,dx + c$ $xy = \sin x - x\cos x + c$	1 1 2 1
45.(a)	$\frac{dx}{dt}\ \alpha\ x \Rightarrow \frac{dx}{dt} = kx \Rightarrow x = ce^{kt}$ $t=0$; $X=x_0 \Rightarrow C=X_0$ $X=X_0e^{kt}$ $t=5$; $X=3x_0 \Rightarrow e^{5K}=3$ $t=10$; $X=9x_0$	1 1 1 1 1
45.(b)	$\vec{a}=2\hat{i}+2\hat{j}+\hat{k}$; $\vec{b}=2\hat{i}+3\hat{j}+3\hat{k}$; $\vec{c}=3\hat{i}+2\hat{j}+\hat{k}$ **Parametric form** $\vec{r}=(2\hat{i}+2\hat{j}+\hat{k})+s(2\hat{i}+3\hat{j}+3\hat{k})+t(3\hat{i}+2\hat{j}+\hat{k})$ **(or) non Parametric form** $\vec{r}.(3\hat{i}-7\hat{j}+5\hat{k})=-3$ **Cartesian form** $\begin{vmatrix} x-2 & y-2 & z-1 \\ 2 & 3 & 3 \\ 3 & 2 & 1 \end{vmatrix}=0$ $3x-7y+5z+3=0$	1 1 2 1
46.(a)	Diagram The required Area $= 4\int_0^a y\,dx = 4\int_0^a \frac{b}{a}\sqrt{a^2-x^2}\,dx$ $= \frac{4b}{a}\left[\frac{x}{2}\sqrt{a^2-x^2}+\frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right)\right]_0^a$ $=\pi ab$ sq.units	1 1 2 1
46.(b)	Equation of parabola:$(y-2)^2=8(x-1)$ Vertex:(1,2) Focus:(3,2) Equation of directrix: x+1=0	2 1 1 1

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47.(a)

p	q	p↔q	∼p	∼q	(∼p)∨q	(∼q)∨p	((∼p)∨q)∧((∼q)∨p)
T	T	T	F	F	T	T	T
T	F	F	F	T	F	T	F
F	T	F	T	F	T	F	F
F	F	T	T	T	T	T	T

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From column (3),(8)

p↔q≡((∼)p∨q)∧((∼)q∨p) 1

47.(b)

Δ= -15 1

ΔX =-15 1

ΔY = -5 1

ΔZ = -5 1

(X,Y,Z)=(1,3,3) 1

MARK ANALYSIS

(WITHOUT CHOICE)

PART	Questions	Total Questions	Book Back Questions	Interior/Creative Questions	Total Marks
I	1 Mark	20	16	4	20
II	2 Marks	10	9	1	20
III	3 Marks	10	9	1	30
IV	5 Marks	14	12	2	70
Total Marks			121	19	140
Percentage			86%	14%	100%

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